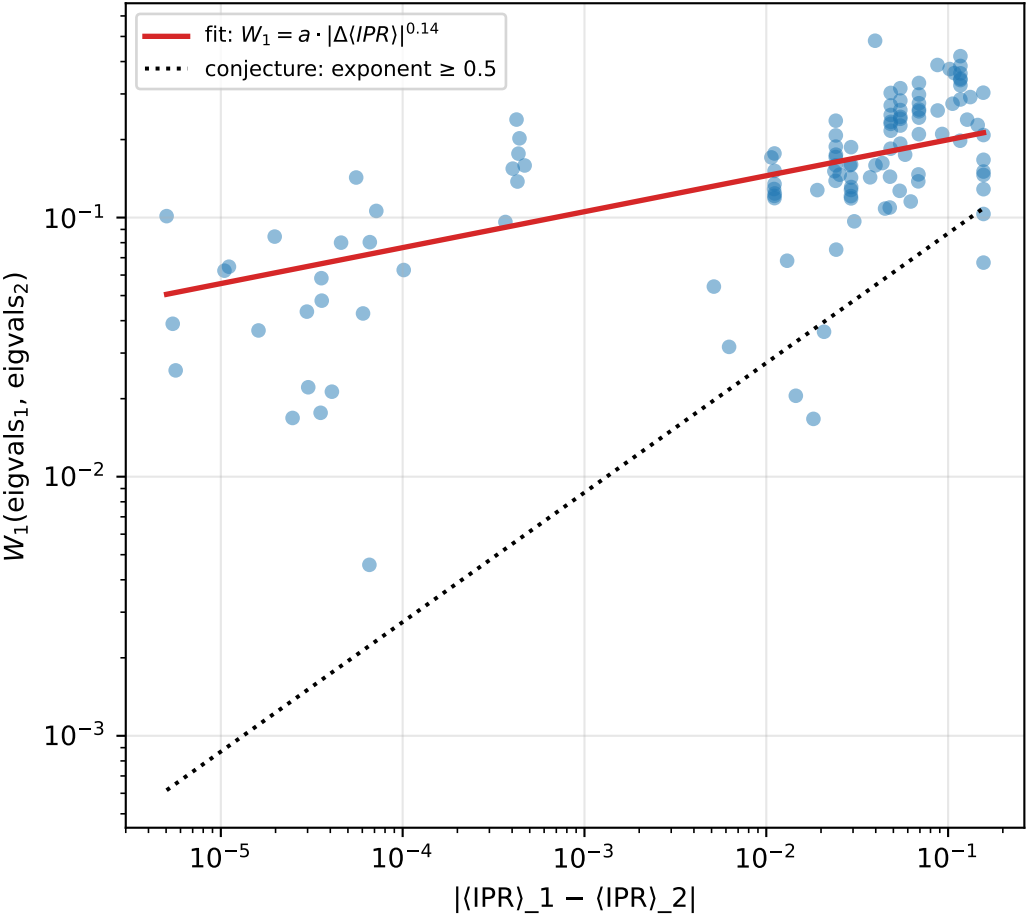
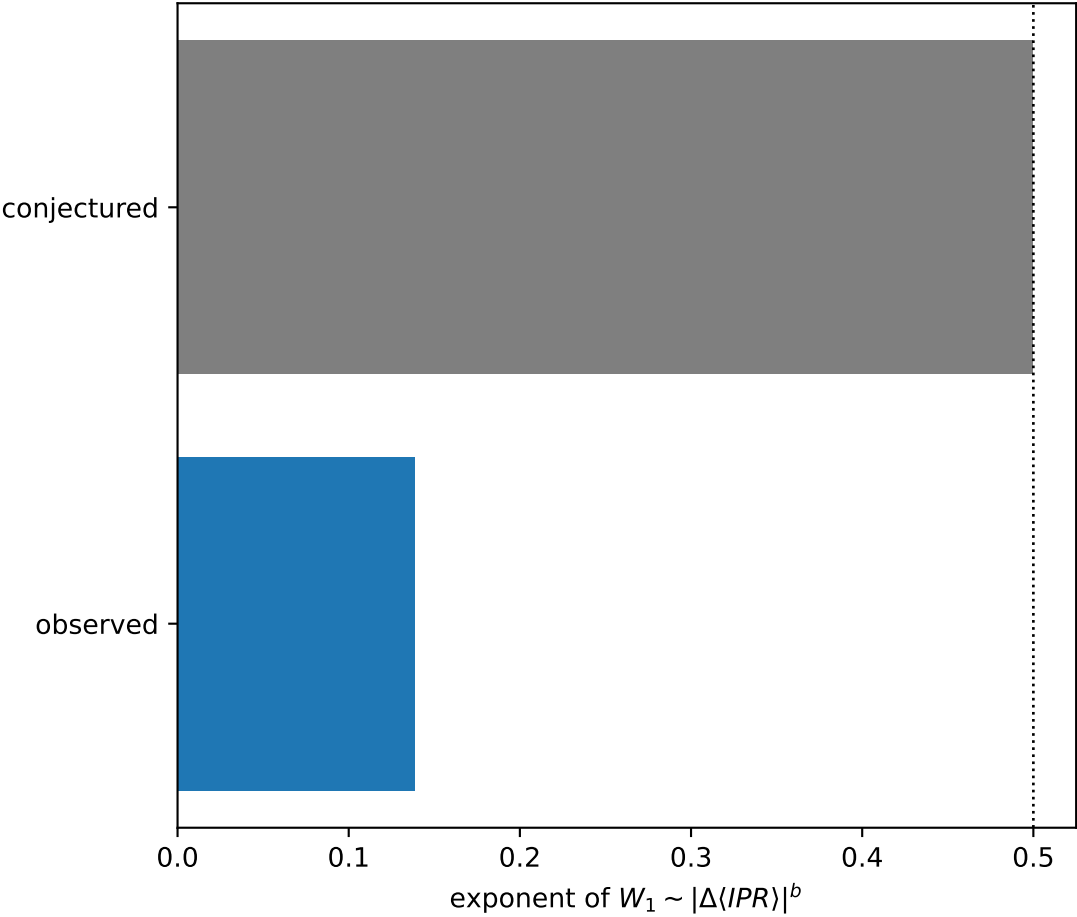


Numerical validation of three mathematical conjectures

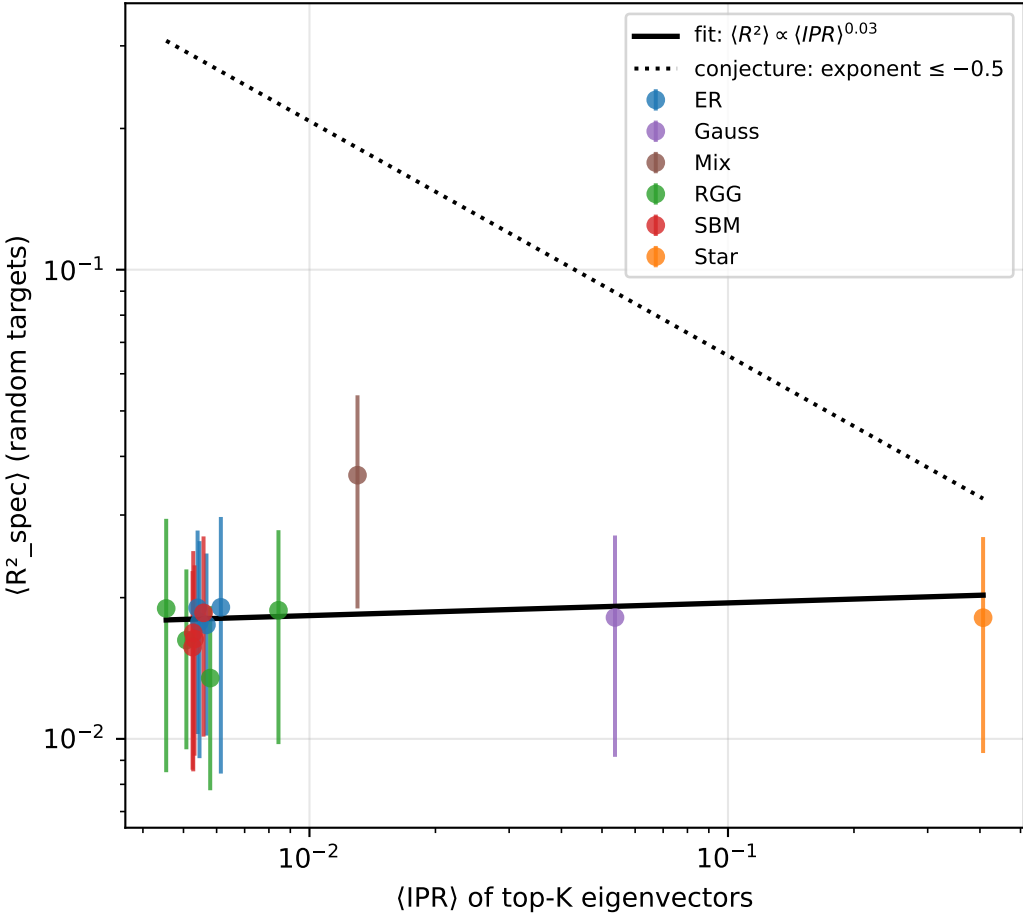
(A) IPR-Wasserstein scaling  
(synthetic ER+RGG+SBM+Gauss)



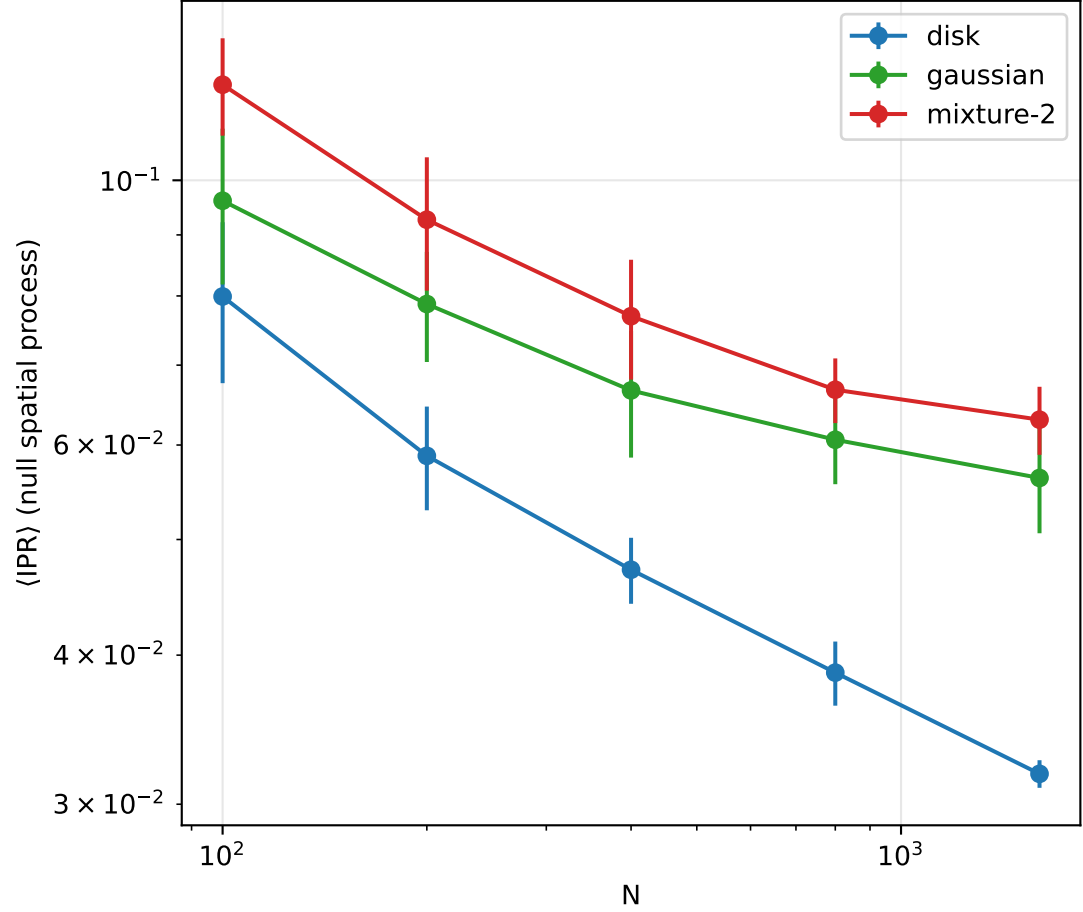
(A) Conjecture A check



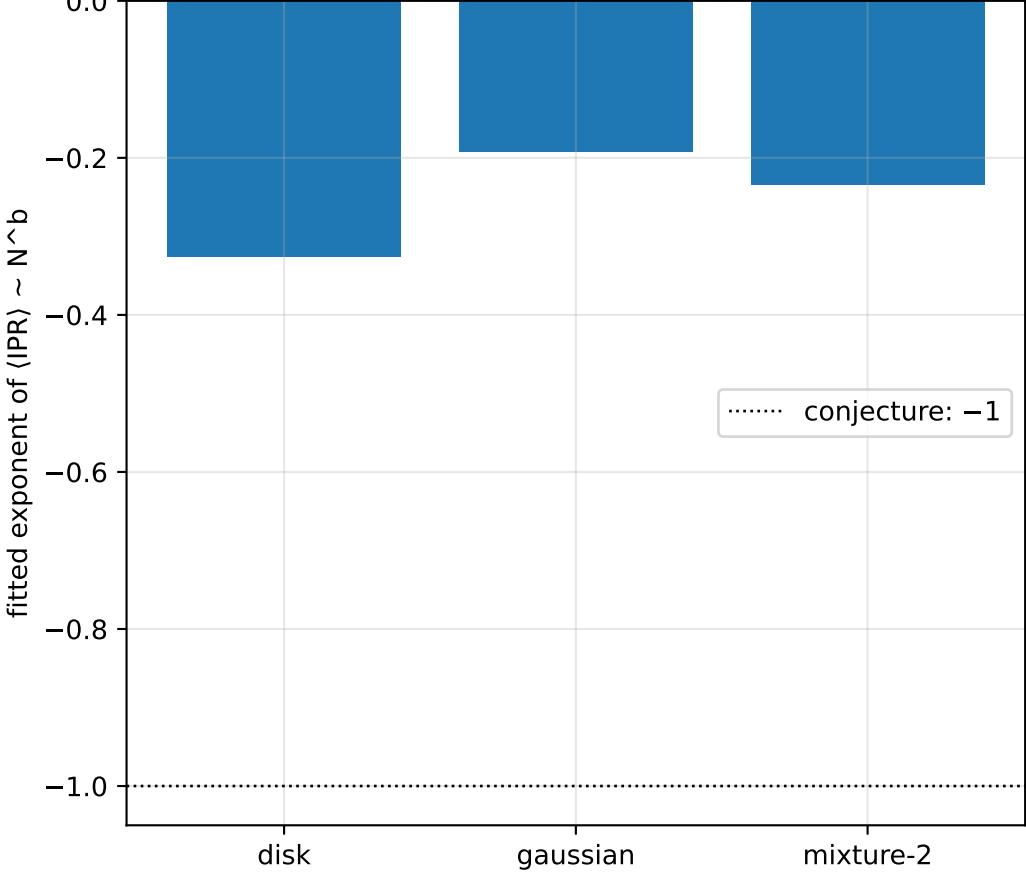
(B) Saturation: more localised  $\rightarrow$  lower  $R^2_{\text{spec}}$  ceiling



(C) Null IPR scales like  $N^{-1}$



(C) Universality check



NUMERICAL VALIDATION SUMMARY

- (A) IPR- $W_1$  exponent: 0.14 (conjectured  $\geq 0.5$ )  
x violates
- (B)  $R^2$ -IPR exponent: 0.03 (conjectured  $\leq -0.5$ )  
x violates
- (C) Null IPR exponent (mean over densities): -0.25  
conjectured: -1  
x violates